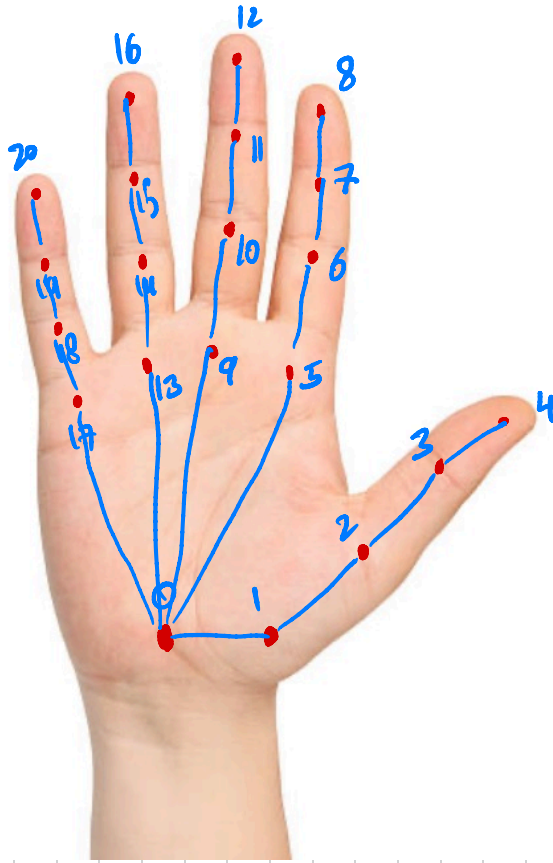




Overall landmarks from pipeline.



each landmark is a 3D point (x, y, z)

x is horizontal position $0-1 \Rightarrow$ left-right

y is vertical position

est. using neural network.

z is depth relative to wrist (how far landmark from camera)

z small $\Rightarrow$ close to cam, large far from cam

$z=0$ same depth as wrist

-0.2 very close to camera

-0.05 slightly close

$+0.2$ very far from camera

$+0.05$ slightly far.

normalization $\Rightarrow$ make wrist (point #0)

as our system origin $(0,0,0)$.

so for each landmark point P we do

$p = p - \text{wrist}$ (including wrist so now its $(0,0,0)$)

now we want to remove scale so hand size matters

less, we do that by dividing by scale which

is the dist between wrist and middle finger (12)

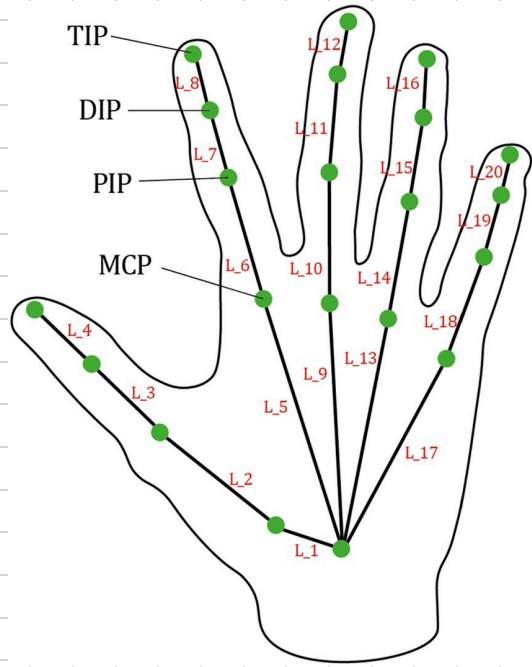
$\text{scale} = \text{landmark}[12].\text{length} \ \& \ p = p / \text{scale}.$

now small scale should be ignored cuz it indicates
issues in recognizing hand position (detection glitch)
or frame noise.

Think about adding upper bounds well to avoid
weird behaviour

Now feature extraction:

the idea is :

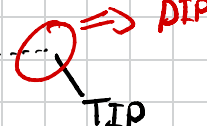


we use the vector of Tip, PIP & MCP that we have to determine if finger is up, using geometry:

$$(TIP - MCP) \cdot (PIP - MCP)$$
 the result will tell us how the 2 vector V_1 & V_2 are in the same direction.

the rule we get $\cos \Theta = \frac{V_1 \cdot V_2}{|V_1| |V_2|}$, the interpretation:

1) MCP - PIP - TIP $\Rightarrow \Theta \approx 0^\circ \Rightarrow \cos \Theta \approx 1$
straight finger

2) MCP - PIP  $\Rightarrow$ $V_1 \rightarrow \Rightarrow \Theta \approx 30-45^\circ$
 $V_2 \rightarrow \Rightarrow \cos \Theta \approx 0.7-0.9$
mostly straight, slightly bent.

3) MCP - PIP $\Rightarrow$ $V_1 \rightarrow$
 $\vdots$ $V_2 \downarrow \Rightarrow \Theta \approx 90^\circ$
 $\Rightarrow \cos \Theta \approx 0$
TIP
bent finger

4) MCP — PIP $\Rightarrow$ $\begin{matrix} V_1 \rightarrow \\ V_2 \swarrow \end{matrix} \Rightarrow \Theta \approx 120-150^\circ$
 $\Rightarrow \cos \Theta = -0.5$
 TIP

finger folded (curled inward).

5) MCP — PIP $\Rightarrow$ $\begin{matrix} V_1 \rightarrow \\ V_2 \leftarrow \end{matrix} \Rightarrow \Theta \approx 180^\circ$
 $\cos \Theta = -1$
 TIP

finger fully folded (like in fist).

after testing decent results Just

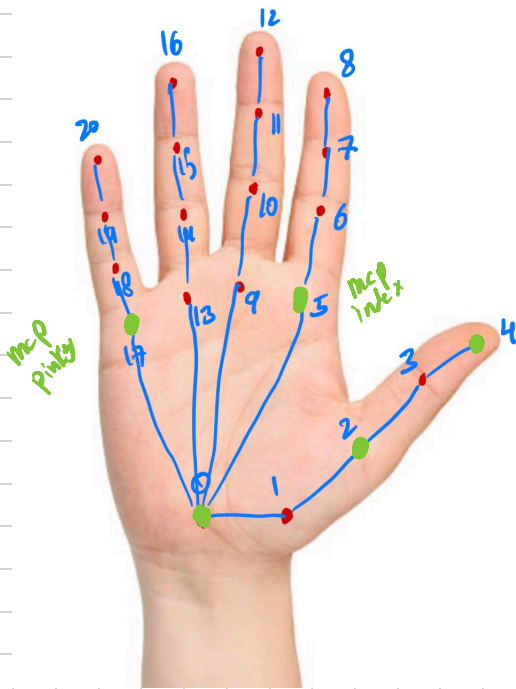
need to think better about ③

when finger is bent (result was inconsistent)

also we thumb participates in fist than 5

becomes a bit unstable -0.9 to -0.6.

Oh well, thump seems to be not working at all, thus after thinking & reading a lil new idea is to deal with it by itself using palm vector.



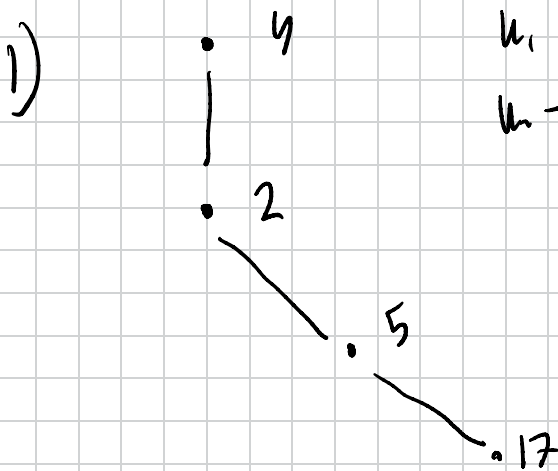
Palm vector = mcp index $\rightarrow$ mcp pinky

this is the direction across the palm

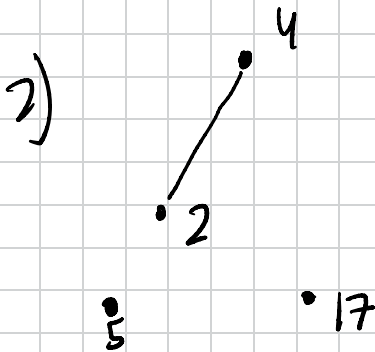
thumb vector still the same
mcp $\rightarrow$ tip of thumb

Now dot product between u_1 palm vec & u_2 thumb vector

after analyzing we get :

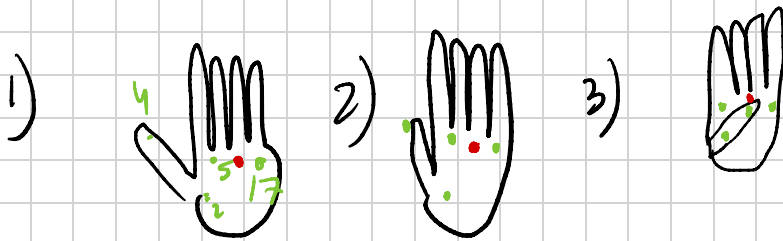


$u_1 \rightarrow \Rightarrow \cos \theta \approx 1$
 $u_2 \rightarrow \Rightarrow$ thumb folded into palm
 (thumb inside palm)



$u_1 \rightarrow \Rightarrow \cos \theta \approx 0$
 $u_2 \uparrow \Rightarrow$ thumbs extended

↳ scrap that, after drawing number on palm
and testing its better to do it on
distance of finger from center of palm,
main 3 option:



↳ Just measure how close is thumb to center •

$$\text{so Center} = \frac{\sqrt{5} + \sqrt{17}}{2}$$

$$\& \text{ diff} = \sqrt{4} - \text{Center}$$

result is $|\text{diff}| \Rightarrow$ extended $\approx > 0.6$

natural $\approx > 0.3$

folded $\approx < 0.3$

* Now new feature for how far finger are apart:

Very simple just do $(V_{\text{Tip}} - V_{\text{Tip}}^*) \cdot \text{length}$ that it..

result ≈ 0.05 touching, 0.15 small gap, 0.3 midgap
 0.5 wide

* and finger direction (relative to wrist)

fairly simple $V_{\text{Tip}} - V_0^{\text{wrist}}$. $1 \leq \text{result} \leq 1$
 x, y, z

$y < -0.6$ up, $y > 0.4$ down, $x > 0.5$ right, < -0.5 left.

* Palm orientation (tilt of hand).

again using ≤ 17 & 0

$$u_1 = V_5 - V_0, \quad u_2 = V_{17} - V_0$$

$$\text{result} = u_1 \times u_2 \Rightarrow x, y, z$$

$z \approx 0 \Rightarrow$ sideways, $z > 0$ Palm facing cam

$z < 0 \Rightarrow$ back of hand facing cam

* dist of finger from wrist (if finger is extended or folded)

ez dist $\text{dist}(V_0, V_{\text{tip}})$

$\approx 0.9 - 1$ fully stretched

≈ 0.7 extended

≈ 0.4 half extended

≈ 0.2 folded

+ Very Cool Interaction between thumb & index
in a lot of sign thus we do thumb, index distance

$(V_4 - V_8) \cdot \text{lengths}$

0.02 touching, 0.1 near, 0.2 in between

0.5 far

need to test more the values of the features
before linking them into actual signs.

Now we build feature vector containing all features

thumb dist

index bend $\rightarrow$ pinky bend

spread IM , spread MR , spread RP
↓ ↓ ↓
index mid mid ring ring, pinky

thumb index dist

index wrist dist

mid wrist dist

palm $\neq$ (orientation).

might add more if need.

Y & X Orientation as well
and added thrust bend.

now I need dataset to for my classifier.

ill do ~ 20 samples per label, we
got A-Z & 0-9 so (ill ignore j & z for now)

those need more work cuz of motion so only

≈ 720 samples, in addition ill inc

the sample size by creating few more by adding

tiny noise like $\text{feature} \pm \text{random}(-0.02, 0.02)$

this should be good.

for now tho ill use the recorded test set "q08"
around 2-30 of each.

now time to chose classifier.